

Thus, our new system of linear equations has the following 4 equations:

$$\left. \begin{array}{l} 3x + y + 2z = 11 \dots (\text{Eq.1}) \\ x + 2y - 2z = -1 \dots (\text{Eq.2}) \\ 2x - 2y - z = -5 \dots (\text{Eq.3}) \\ 4x - 3y + 2z = -2 \dots (\text{Eq.5}) \end{array} \right\} \dots (\text{H})$$

By solving first three equations we get $x=1$, $y=2$, and $z=3$. If we substitute these values in (Eq. 5), we get $\text{LHS} = 4(1) - 3(2) + 2(3) = 4 \neq -2$, the RHS.

It can be checked that solution of any of 3 equations out of the 4 equations in (H), does not satisfy the remaining 4th equation.

Such set of $(n+1)$ or more equations in n variables, is called ***inconsistent system of equations***.

However, the above statement does not imply that every set of $(n+1)$ or more equations in n variables must be inconsistent.

For example, instead of (Eq. 5), we add the following equation to (C)

$$4x - 3y + 2z = 4 \dots (\text{Eq. 6}),$$

to get a new system of linear equations

$$\left. \begin{array}{l} 3x + y + 2z = 11 \dots (\text{Eq.1}) \\ x + 2y - 2z = -1 \dots (\text{Eq.2}) \\ 2x - 2y - z = -5 \dots (\text{Eq.3}) \\ 4x - 3y + 2z = 4 \dots (\text{Eq.6}) \end{array} \right\} \dots (\text{L})$$

Then, all the four equations of (L) are simultaneously satisfied by

$$x=1, y=2, \text{ and } z=3.$$

Such a system of linear equations is called a CONSISTENT system of equations.

Definitions:

- i) A system of m linear equations in n variables, with $m > n$, is said to be ***consistent***, if these equations have a solution, like equations in (L) above.
- ii) A system of m linear equations in n variables, with $m > n$, is said to be ***inconsistent***, if these equations do not have any (common) solution, like equations in (H) above.

Remarks:

- i) A system of m linear equations in n variables, with $m \leq n$, is necessarily/ always consistent. The question of consistency/ inconsistency arises only when $m > n$.
- ii) Any set of $(n+1)$ or more linear equations in n variables is either consistent system or it is inconsistent system. If system is consistent, then it must be

linearly dependent system of equations. If it is inconsistent, then it is linearly independent.

(A) Test for consistency of m equations in n variables, with $m > n$.

The augmented matrix, for a system of m equations in n variables is $m \times (n+1)$ matrix, where

$$m \geq (n+1).$$

If through elementary row operations, the $(n+1)^{th}$ row (or any other row) can be reduced to $(0, 0, \dots, 0, K)$, (or $(0, 0, \dots, K, 0, \dots, 0)$ in which any n entries are all 0's, and $K \neq 0$, then the system of equations is INCONSISTENT.

The system is inconsistent, because the row $(0, 0, \dots, 0, K)$ implies

$$0 = 0x_1 + 0x_2 + \dots + 0x_n = K \neq 0.$$

On the other hand, if through elementary row operations, whenever, we get a **row of the form $(0, 0, \dots, 0, K)$, then $K=0$ implies** then the system of m linear equations in n variables, with $m > n$, is a **consistent system**.

Examples

i) *Inconsistent system of equations*

The augmented matrix corresponding to the system of equations (H) is

$$\left[\begin{array}{ccc|c} 3 & 1 & 2 & 11 \\ 1 & 2 & -2 & -1 \\ 2 & -2 & -1 & -5 \\ 4 & -3 & 2 & -2 \end{array} \right]$$

Like the various elementary row operations used for obtaining E^{vii} , we may get

$$\left[\begin{array}{ccc|c} 1 & 2 & -2 & -1 \\ 0 & -5 & 8 & 14 \\ 0 & 0 & 1 & 3 \\ 4 & -3 & 2 & -2 \end{array} \right]$$

By $R_1 \rightarrow 3R_1 + R_2$, we get

$$\left[\begin{array}{ccc|c} 3 & 1 & 2 & 11 \\ 0 & -5 & 8 & 14 \\ 0 & 0 & 1 & 3 \\ 4 & -3 & 2 & -2 \end{array} \right]$$

By $R_4 \rightarrow 3R_4 - 4R_1$, we get

$$\left[\begin{array}{ccc|c} 3 & 1 & 2 & 11 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 1 & 3 \\ 0 & -13 & -2 & -50 \end{array} \right]$$

By $R_4 \rightarrow R_4 + 13R_2$, we get

$$\left[\begin{array}{ccc|c} 1 & 2 & -2 & -1 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 24 & 54 \end{array} \right]$$

By $R_4 \rightarrow R_4 - 24 R_3$, we get

$$\left[\begin{array}{ccc|c} 1 & 2 & -2 & -1 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & -18 \end{array} \right]$$

The last row implies

$$0 = 0x_1 + 0x_2 + 0x_3 = -18, \text{ which is not possible.}$$

Hence, the system is inconsistent.

ii) *Consistent system of equations.*

The following system of equations (L)

$$\left[\begin{array}{ccc|c} 3 & 1 & 2 & 11 \\ 1 & 2 & -2 & -1 \\ 2 & -2 & -1 & -5 \\ 4 & -3 & 2 & 4 \end{array} \right]$$

is quite similar to the previous one of (H), except entry (4th row, 4th column) are different.

Hence, we apply exactly the same sequence of elementary operations. Prior to the last elementary operation, we get

$$\left[\begin{array}{ccc|c} 1 & 2 & -2 & -1 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 1 & 3 \\ 4 & -3 & 2 & 4 \end{array} \right]$$

And, by $R_4 \rightarrow R_4 - 4 R_1$, we get

$$\left[\begin{array}{ccc|c} 1 & 2 & -2 & -1 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 1 & 3 \\ 0 & -11 & 10 & 8 \end{array} \right]$$

$R_4 \rightarrow R_4 + 11 R_2$, we get

$$\left[\begin{array}{ccc|c} 1 & 2 & -2 & -1 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 32 & 96 \end{array} \right]$$

Finally, $R_4 \rightarrow R_4 - 32R_3$, we get

$$\left[\begin{array}{ccc|c} 1 & 2 & -2 & -1 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

None of the rows is of the form

$(0, 0, \dots, 0, K)$ with $K \neq 0$.

Hence the system (L) is consistent.

8.7 RANK OF A MATRIX AND SOLUTION OF A SYSTEM OF EQUATIONS

The concept of 'Rank of a Matrix', and some related issues have been already discussed under section 5.3.7 of Unit 5.

Here we just state some results connecting ranks of matrices A and augmented matrix $[A|B]$ to the number of solutions, and consistency of a system of m equations in n variables, represented by the matrix equation. Let

$$AX = B, \quad \text{then}$$

- 1) If rank of matrix $A = \text{rank } [A|B] = m$, then the system has a unique solution.
- 2) If rank of matrix $A = \text{rank } [A|B] < m$, then the system has infinite solutions.
- 3) If rank of matrix $A < \text{rank } [A|B]$, then the system is inconsistent, i.e., the system has no solution

Examples

- i) For the system of equations (C), considered earlier,

$$\left. \begin{array}{l} 3x + y + 2z = 11 \dots (\text{Eq.1}) \\ x + 2y - 2z = -1 \dots (\text{Eq.2}) \\ 2x - 2y - z = -5 \dots (\text{Eq.3}) \end{array} \right\} \dots (C)$$

the augmented matrix can be reduced, through elementary row operations, to

$$\left[\begin{array}{ccc|c} 1 & 2 & -2 & -1 \\ 0 & -5 & 8 & 14 \\ 0 & 0 & 1 & 3 \end{array} \right] \dots (E^{vii})$$

In which the $\text{Rank}(A) = \text{Rank } [A|B] = 3$, the number of equations.

We have already shown that the system has unique solution $x=1$, $y=2$, and $z=3$.

- ii) for 'If rank of matrix $A = \text{rank } [A|B] < m$, then the system has infinite solutions'

Consider

$$\left. \begin{array}{l} 3x + y + 2z = 11 \dots (\text{Eq.1}) \\ x + 2y - 2z = -1 \dots (\text{Eq.2}) \\ 9x + 8y - 2z = 19 \dots (\text{Eq.4}) \end{array} \right\} \dots (F)$$

Its augmented matrix can be reduced to (as done earlier)

$$\left[\begin{array}{ccc|c} 3 & 1 & 2 & 11 \\ 1 & 2 & -2 & -4 \\ 0 & 0 & 0 & 0 \end{array} \right] \dots (G')$$

Here, the rank $(A) = \text{rank of } [A|B] = 2 < 3$, the number of equations. The corresponding system has infinitely many solutions.

Check Your Progress 2

- 1) For each of the following system of equations, write corresponding matrix equation, and augmented matrix. Further, check for linear independence/dependence and/or consistency/inconsistency, whichever is relevant:

(i) $2x = 6$ $3y = 9$	(ii) $3x + 2y = 11$ $-2x + 3y = -29$ $x + y = 2$	(iii) $3x + 2y = 11$ $x + y = 2$ $2x - y = 8$
(iv) $x + 2y = 5$ $-2x + 3z = -11$ $y - 2z = 8$ $x + y + z = 0$	(v) $x + 2y = 5$ $-2x + 3z = -11$ $y - 2z = 8$ $2x + y - z = 12$	(vi) $x + 2y = 5$ $-2x + 3z = -11$ $y - 2z = 8$

.....
.....
.....

8.8 SPAN OF A SET OF VECTORS, BASIS AND DIMENSION IN $\mathbb{R}^N$

Coming back to our earlier example of system of equations (C)

$$\left. \begin{aligned} 3x + y + 2z &= 11 \dots (\text{Eq.1}) \\ x + 2y - 2z &= -1 \dots (\text{Eq.2}) \\ 2x - 2y - z &= -5 \dots (\text{Eq.3}) \end{aligned} \right\} \dots (C)$$

This system was considered as represented by the augmented matrix

$$\left[\begin{array}{ccc|c} 3 & 1 & 2 & 11 \\ 1 & 2 & -2 & -1 \\ 2 & -2 & -1 & -5 \end{array} \right] \dots (E)$$

where, each row represents an equation of (C).

Each row may be independently considered to be represented by the vector

(Coefficient of x, coefficient of y, coefficient of z, constant on RHS).

Thus, a set of equations, in n variables, can equivalently be considered as a set of (n+1)-vectors

For example, the set of equations, in 3 variables, in (C) can be represented as three 4-tuples or 4-vectors

$$V_1 = (3 \ 1 \ 2 \ 11)$$

$$V_2 = (1 \ 2 \ -2 \ -1) \quad \text{and}$$

$$V_3 = (2 \ -2 \ -1 \ -5)$$

where, each of the above vectors V_j is an element of $\mathbb{R}^4 = \mathbb{R} \times \mathbb{R} \times \mathbb{R} \times \mathbb{R}$.

All the **definitions about equations**, like linear combination, linearly dependent and linearly independent, equations **can now be carried over to vectors** e. g. linear combination of vectors, linearly dependent set of vectors and linearly independent set of vectors.

Now, we consider correspondence of system of equations in $(n - 1)$ variables and vectors of length n , or equivalently n -tuples, i. e. vectors in the R^n vector space.

A) Linear combination of vectors:

A vector V is said to be linear combination of vectors $V_1, V_2, \dots, V_k$ if there exist constants λ_j such that

$$V = \lambda_1 V_1 + \lambda_2 V_2 + \dots + \lambda_j V_j + \dots + \lambda_k V_k \text{ for } j = 1, 2, \dots, k.$$

B) Linearly dependent set of vectors:

A set of vectors $V_1, V_2, \dots, V_k$ is said to be linearly dependent, if there exist constants λ_j , not all zeros, such that

$$\lambda_1 V_1 + \lambda_2 V_2 + \dots + \lambda_j V_j + \dots + \lambda_k V_k = \mathbf{0},$$

where $\mathbf{0}$ on RHS denotes zero-vector.

C) Linearly independent set of vectors:

A set of vectors $V_1, V_2, \dots, V_k$ is said to be linearly independent, if the following vector equation

$$\lambda_1 V_1 + \lambda_2 V_2 + \dots + \lambda_j V_j + \dots + \lambda_k V_k = \mathbf{0},$$

is possible only for $\lambda_j = 0$, for all $j = 1, 2, \dots, k$.

The procedure for showing a set of vectors as linearly dependent/ independent is the same as that for showing a set of equations as linearly dependent/ independent.

Span of k vectors: for given k vectors $V_1, V_2, \dots, V_k$ its linear span is the set of all vectors of the following form

$$\lambda_1 V_1 + \lambda_2 V_2 + \dots + \lambda_j V_j + \dots + \lambda_k V_k,$$

where each of λ is a real number.

Span of a set of vectors is called **vector space generated by** the set of vectors.

Basis of a vector space is a set of its vectors which is maximal linearly independent set in the vector space.

Dimension of a vector space is the number of vectors in the basis of the vector space.

8.9 LET US SUM UP

In this unit, first we explained what is meant by a system of linear equations, and what is meant by its solution. Then we explained how to represent a system of linear equations as matrix equation. Further, we defined the concepts of elementary row operations and how to use these operations to get a solution for corresponding system of equations.

Next, we explained the following concepts (i) Linearly independent/dependent system of equations, and (ii) Consistency/ inconsistency of system of m linear equations in n variables, when $m > n$.

Further, we discussed how to use the rank of a matrix for determining linear independence / dependence and consistency/inconsistency of a system of linear equations. Finally, as an optional reading, we explained the concepts of span of a set of vectors, basis and dimension in $\mathbb{R}^n$.

8.10 KEY WORDS

Augmented Matrix : For a given system of linear equations, a matrix formed by taking only the constants of each of the equation as a row, is called the augmented matrix of the given system of equations. For example, for the set of linear equations

$$3x + y + 2z = 11 \dots (\text{Eq.1})$$

$$x + 2y - 2z = -1 \dots (\text{Eq.2})$$

$$9x + 8y - 2z = 19 \dots (\text{Eq.3})$$

its augmented matrix is

$$\left[\begin{array}{ccc|c} 3 & 1 & 2 & 11 \\ 1 & 2 & -2 & -1 \\ 9 & 8 & -2 & 19 \end{array} \right]$$

Consistent system of equations

A system of m linear equations in n variables, with $m > n$, is said to be **consistent**, if these equations have a solution.

Inconsistent system of equations

A system of m linear equations in n variables, with $m > n$, is said to be **inconsistent**, if these equations do not have any (common) solutions.

Linear Equation

: An equation, in which each of the occurring variables has degree not exceeding one, at least one variable has degree one, and coefficient of at least one variable with degree one, is non-zero.

Linear combination

: An equation, say (Eq. k), is said to be **linear combination** of equations (Eq. 1), (Eq. 2),

Linearly Dependent system of equations

Eq. (k-1)), (Eq. (k+1)), ... (Eq. n), if the **equation (Eq. k) can be represented as** $(\text{Eq. } k) = \lambda_1 (\text{Eq. } 1) + \lambda_2 (\text{Eq. } 2) + \dots + \lambda_{(k-1)} (\text{Eq. } (k-1)) + \lambda_{(k+1)} (\text{Eq. } (k+1)) + \lambda_n (\text{Eq. } n)$, for constants λ_j .

A set of m equations, say, (Eq. 1), (Eq. 2), ... & (Eq. m) in n variables, where $m \leq n$, is said to be *linearly dependent* if, for some j with for $1 \leq j \leq m$, the equation, say (Eq. j) can be written as a linear combination of the other equations.

Linearly independent system of equations

: A set of m equations, say, (Eq. 1), (Eq. 2), ... & (Eq. m) in n variables, where $m \leq n$, is said to be *linearly independent* if, for **no** j with for $1 \leq j \leq m$, the equation, say (Eq. j) can be written as a linear combination of the other equations.

Solution of a linear equation

: $a_1 x_1 + a_2 x_2 + \dots + a_n x_n = b \dots \dots \dots (I)$ is a set of n constant real/ complex values c_j for $j = 1, 2, \dots, n$ such that when the variable x_j is replaced by the corresponding constant c_j in equation (I), then value of LHS becomes the constant b , which is the RHS of (I).

System of Linear Equations

: A collection of m linear equations in n variables, where the same values of variables are sought to satisfy all the equations simultaneously.

Linear combination of vectors

: A vector V is said to be linear combination of vectors $V_1, V_2, \dots V_k$ if there exist constants λ_j such that $V = \lambda_1 V_1 + \lambda_2 V_2 + \dots + \lambda_j V_j + \dots + \lambda_k V_k$ for $1 \leq j \leq k$.

Linearly dependent set of vectors

: A set of vectors $V_1, V_2, \dots V_k$ is said to be linearly dependent, if there exist constants λ_j , not all zeros, such that

$$\lambda_1 V_1 + \lambda_2 V_2 + \dots + \lambda_j V_j + \dots + \lambda_k V_k = 0,$$

where 0 on RHS denotes zero-vector.

Linearly independent set of vectors

: A set of vectors $V_1, V_2, \dots V_k$ is said to be linearly independent, if the following vector equation

$$\lambda_1 V_1 + \lambda_2 V_2 + \dots + \lambda_j V_j + \dots + \lambda_k V_k = 0,$$

is possible only for $\lambda_j = 0$, for all $j = 1, 2, \dots, k$.

8.11 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1)
- i) $3x + 4y = 0$ is linear equation
 - ii) $2x - 11$, is not even an equation
 - iii) $x y = 3$, is not a linear equation
 - iv) $7 + 4 = 11$ is not a linear equation, as it does not involve any variable
 - v) $x \cdot (x - 1) = 0$, is not linear, as in it, exponent of x is 2 also.
- 2)
- i) $x(x + 2) = 0$ is not even a linear equation
 - ii) $x - 4 + y = 0$, is linear, but not homogeneous
 - iii) $x + 3y + 4z = 0$, is homogenous linear equation.

Check Your Progress 2

1) i) $\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 9 \end{bmatrix}$

Augmented matrix =

$$\left[\begin{array}{cc|c} 2 & 0 & 6 \\ 0 & 3 & 9 \end{array} \right]$$

The system of equations is linearly independent, as the rank of augmented matrix is 2.

For the remaining parts, we give only augmented matrix, and not matrix equation.

ii) $\left[\begin{array}{cc|c} 3 & 2 & 11 \\ -2 & 3 & -29 \\ 1 & 1 & 2 \end{array} \right]$ (iii) $\left[\begin{array}{cc|c} 3 & 2 & 11 \\ 1 & 1 & 2 \\ 2 & -1 & 8 \end{array} \right]$

For (ii) As number of equations $= 3 > 2 =$ number of variables. Issue of consistency has to be checked. As $x = 7$, $y = -5$ satisfies all the equations, the system is consistent, and hence a linearly dependent system of equations.

For (iii) Though $x = 7$, and $y = -5$ satisfies first two equations, it does not satisfy (iii), the system is inconsistent, and hence, it is a linearly independent system of linear equations.

iv) $\left[\begin{array}{ccc|c} 1 & 2 & 0 & 5 \\ -2 & 0 & 3 & -11 \\ 0 & 1 & -2 & 8 \end{array} \right]$

v) $\left[\begin{array}{ccc|c} 1 & 2 & 0 & 5 \\ -2 & 0 & 3 & -11 \\ 0 & 1 & -2 & 8 \\ 1 & 1 & 1 & 0 \end{array} \right]$

As all the equations are satisfied by $x = 1$, $y = 2$ and $z = -3$, hence the system is consistent. Also, number of equations $>$ number of variables, a consistent system is linearly dependent.

$$\text{vi)} \quad \left[\begin{array}{ccc|c} 1 & 2 & 0 & 5 \\ -2 & 0 & 3 & -11 \\ 0 & 1 & -2 & 8 \\ 1 & 1 & 1 & 12 \end{array} \right]$$

As the first three equations are satisfied by $x = 1$, $y = 2$ and $z = -3$, but NOT the last. Hence system is inconsistent.

8.12 EXERCISES

Q1. Define each of the following concepts, and give one relevant example as illustration for each

a) Linearly independent system of equations

Ans. The following two equations are linearly independent

$$X=3; \text{ and } 2X+Y=7$$

b) Linearly dependent system of equations

Ans. The following two equations are linearly dependent

$$X+2Y=6, \text{ and}$$

$$2X+4Y=12$$

c) Consistent system of equations

Ans. The following system of equations

$$X+2Y=8,$$

$$3X+4Y=18$$

$$5X+3Y=19$$

is a consistent system of 3 equations in 2 variables, as it has a solution $X=2$, and $Y=3$.

d) Inconsistent system of equations

Ans. The following system of equations

$$X+2Y=8,$$

$$3X+4Y=18$$

$$5X+3Y=20$$

is an inconsistent system of 3 equations in 2 variables, as it does NOT have any solution.